Optimizing Operational Efficiency and Seasonal Profitability: A Data-Driven Strategy for an Air Conditioner Repair Center

A Proposal report for the BDM capstone Project

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Contents –

Declaration Statement

I am working on a Project Title "**Optimizing Operational Efficiency and Seasonal Profitability: A Data-Driven Strategy for an Air Conditioner Repair Center**". I extend my appreciation to **Aone Cooling Centre**, for providing the necessary resources that enabled me to conduct my project.

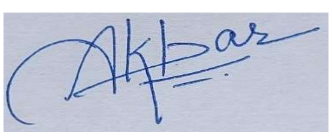
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.



Signature of Candidate:

Name: Akbar Ali

Date: 03/02/2025

1. Executive Summary –

This project is focusing on a medium sized A.C. Repairing Center **i.e. Aone Cooling Center**, a business to customer service-based business that specializes in the maintenance and repair of air conditioning units. The center plays a very important role in providing timely and efficient services to customers, ensuring optimal performance of their air conditioning systems. They have also started a small business side by side that is of selling new A.C. to the customers but the main source of profit for them is by repairing the damaged air conditioning systems.

Besides its importance, the business faces some significant challenges that impact operational efficiency and profitability. The main issues among them are somehow related to the buying costs and worker management strategies with the variation in the season and changes in the weather. The major issues will most probably be addressed after analyzing historical service and procurement data to identify price trends, seasonal fluctuations, and workforce efficiency patterns.

The business can optimize by managing the timing of buying spare parts and materials to develop cost-effective buying strategies and controlling the number of workers that are being hired to enhance overall operational efficiency and profitability. This project aims to provide practical recommendations that can be implemented to improve decision-making processes and long-term sustainability.

2. Organization Background –

The **Aone Cooling Centre** is a service provider that is dedicated towards ensuring the efficient operation and maintenance of air conditioning systems. With a strong presence in the industry, the business has built a good reputation for delivering **long-lasting repair and maintenance services**, catering mostly to residential, commercial, and a little to industrial clients also.
The organization with workers having almost 15 years of experience specializes in **diagnostics, routine maintenance, repairs, and part replacements**, ensuring optimal cooling performance and energy efficiency. One of the key aspects of its operations is maintaining a skilled workforce of technicians who are well-trained in handling various AC models and brands.
The business also focuses on **procurement and inventory management**, ensuring the availability of all essential spare parts to minimize the days of repairing and servicing. With a commitment to **customer satisfaction, operational efficiency, and cost-effectiveness**, the repair center continuously tries its best to enhance service quality. It aims to optimize **workforce management and procurement planning**, addressing industry challenges and ensuring sustainable growth in a competitive market.

3. Problem Statements –

3.1 Problem Statement 1: Procurement Cost Management and Inventory Optimization

We need to implement data-driven procurement strategies to analyze spare parts demand and analyze historical price trends. This will help reduce excess inventory costs, as well as prevent stock shortage, and ensure cost-effective purchasing during different seasons.

3.2 Problem Statement 2: Workforce Optimization for Efficient Service Delivery

We need to develop a data-driven workforce management strategy to optimize the number of hired workers based on seasonal demand. This will help manage service delays, minimize idle labor costs during off seasons, and ensure consistent and effective service quality throughout the year.

4. Background of the Problem –

Managing a repair center involves more than just fixing machines—it requires hired worker management and strategic procurement planning too. However, two major challenges hinder smooth operations: hiring required workers only and manage fluctuating purchasing costs. These factors directly impact service efficiency, financial stability, and overall business performance.

Main Causes for the occurrence of the problem are:

- The prices of spare parts and materials that are used vary with seasonal demand. Purchasing in excess ties up capital, while under-purchasing may cause service disruptions.
- Workload is uneven in different seasons. Peak seasons lead to service delays due to lack of workers, while off-seasons result in underutilized workers and unnecessary labor costs.

Challenges from the Inside

Actually, there is an absence of structured data tracking that creates roadblocks. Theirs decisions are mostly based on experience instead of real-time analytics, making it difficult to adapt quickly to changing demands.

External Pressures

The market is unpredictable. Seasonal demand spikes, supply chain fluctuations, and rising customer expectations make it hard to maintain steady operations. One wrong move in inventory or workforce management could lead to financial strain and lost opportunities.

To stay competitive, the center needs a smart, data-driven strategy. Analyzing historical trends and predicting future needs will not only streamline operations but also reduce costs, improve service efficiency, and ensure better workforce utilization. By addressing these challenges head-on, the business can turn unpredictability into opportunity and establish itself as a leader in the industry.

5. Problem Solving Approach

1. **Methods Used in Solving the Problems:**
   To optimize cost management and reduce repair delays, these methods provide a structured approach.

- **Optimized Inventory Management**: Maintaining a database for spare parts availability and track those that are either in excess or in shortage in both season and no seasonal weather. Utilize historical sales data to identify trends and patterns, enabling data-driven decision-making.
- **Demand Forecasting for Workforce Management:** Efficient management of number of hired workers is essential for minimizing service delays and cost. Analytical tools such as workforce utilization analysis, turnaround time metrics, and customer satisfaction trends will be used to optimize resource allocation and controlling them in numbers. These methods will help maintain service reliability while avoiding excessive staffing costs during low-demand periods.

2. **Data Collection Requirements:**
   To make fruitful decisions, accurate and relevant data will be gathered through:

- **Inventory and Procurement Data**: Tracking stock availability and parts usage patterns in different seasons to streamline inventory management and also get rid of those that are stored for long and hold capital.
- **Cost Analysis Reports**: Evaluating expenses on workers in relation to peak seasons and normal days to maintain profitability. Also, a record of number of workers required in all seasons.
- **Customer Feedback & Complaint Reports**: Understanding customer satisfaction levels and identifying areas requiring service improvements.
  This data will support strategic workforce planning, ensuring optimal staffing levels while maintaining service quality and cost-effectiveness.

3. **Analysis Tools:**
   In order to derive actionable insights from the collected data, the following analytical tools will be utilized:

- **Inventory Turnover Analysis**: Measuring stock movement to balance supply, demand and excess of spare parts.
- **Trend and Predictive Analysis**: Using statistical models to identify seasonal repair patterns and future demand forecasts, ensuring resource availability. Also clearing the excess stock.

- **Workforce Demand Analysis:** Using predictive modeling to anticipate workload surges and plan staffing accordingly. Analyzing worker-to-service ratios to balance worker costs with revenue generation.
- **Customer Sentiment Analysis**: Analyzing customer reviews and ratings using qualitative analysis to enhance service quality.

  These tools will provide actionable insights to optimize operations, reduce service delays, and improve customer satisfaction.

6. Work Breakdown and Expected Timeline –

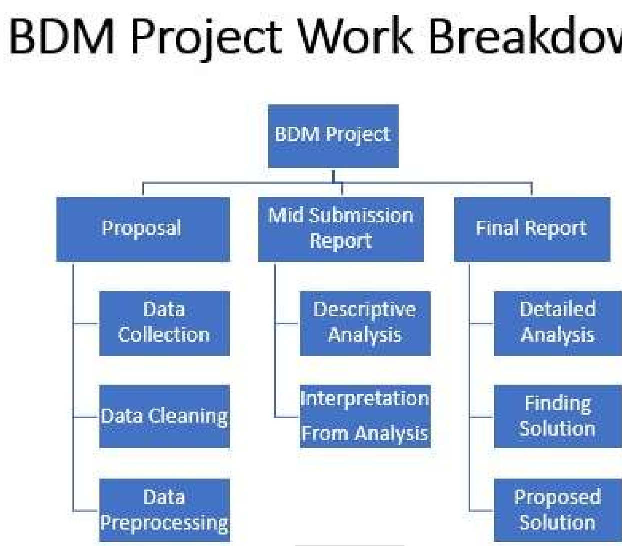


Fig 1

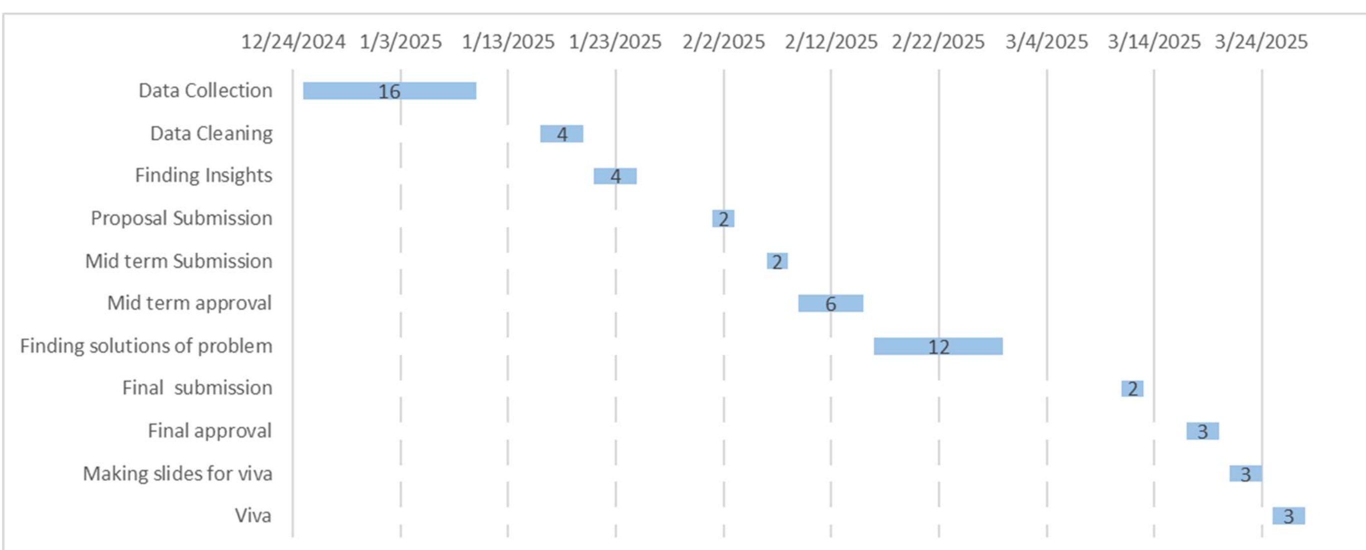


Fig 2: Gantt Chart

7. Expected Outcome –

Improved Inventory Planning: Tracking the usage of spare parts will help prevent overstocking, reduce costs, and ensure that parts are available when needed.

Cost Control and Profitability: Optimizing workforce hiring and managing expenses will increase financial efficiency and profitability.

Optimized Workforce Management: Efficiently allocating workers based on demand will lower costs and enhance service efficiency.

Enhanced Customer Satisfaction: Providing faster service and improving quality will strengthen customer trust and loyalty.